



Impact of cyanobacterial bloom on microbiomes of freshwater lakes

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Cyanobacterial blooms are harmful because of their cyanotoxins production. It occurs due to the eutrophication of freshwater reserves. Nagpur city has three lakes which serves as public water resource and are affected by algal bloom events. Metagenomic approach was used for the exploration of taxonomic, catabolic, and resistome profile of these lakes. Taxonomic profiling indicated prevalence of cyanobacterial species like *Microcystis*, *Anabaena*, *Trichodesmium*, *Microcoleus*, and *Nodularia*. These bacteria are well known for their association with algal bloom and microcystin production. High Performance Liquid Chromatography (HPLC) detected the presence of microcystin toxin and concentration was relatively higher in Ambazari lakewater. In addition, the presence of several antibiotic resistance genes like vancomycin, beta-lactamase, methicillin, and fluoroquinolones were observed. Genes conferring metal resistance such as copper, cadmium, zinc were also mined indicating presence of metal resistant microorganisms. The study suggests that lake water contaminated with *Microcystis* (algal bloom) harbors complex microbial community having diverse catabolic and resistome profiles, which negatively affect the ecosystems services.

Keywords. Cyanobacteria; *Microcystis*; Metagenomics; Gene; Toxicity; Antibiotic resistance

1. Introduction

Cyanobacteria are ubiquitous and have played significant roles in geochemical events such as the 'evolution of eukaryotes and multicellular organisms' through oxygenation of the atmosphere (Hofer 2013). Cyanobacteria has the potential to form compact and sometimes toxic blooms in organic-rich marine and freshwater environments (Huisman *et al.* 2018). An algal bloom may be defined as the rapid accumulation or increase of algal bacteria in aquatic ecosystems (freshwater or marine water). The term algae include both macroscopic, multicellular organisms like seaweed and microscopic, unicellular photosynthetic organisms like cyanobacteria. Environmental conditions that favour algal bloom growth tend to occur

in late summer and autumn in temperate zones and potentially year-round in productive tropical and subtropical zones. Also, global climate change mediated by rise in temperature, and organic enrichment plays a key role in the episodes of algal bloom. It has been suggested that ocean acidification alters marine organism's physiology and induces toxic algal bloom (Riebesell *et al.* 2018). It also changes the dynamics of the microbial biosphere by decreasing the relative fitness of microbes via altering the cellular processes, including acid-base regulation, nutrient uptake, and calcification, and thus favors the proliferation of harmful algal species. Algal blooms also cause oxygen depletion affecting aquatic biodiversity and nutrient recycling. This spread of algal blooms imposes threat to both ecosystem and human health globally by the

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production of cyanobacterial toxins (Pal et al. 2021). These toxins are hepatotoxic and neurotoxic (Deng et al. 2017). Microcystins (MCs) are the commonly found cyanotoxins in algal bloom contaminated water (Sangolkar et al. 2009; Pal et al. 2020). They are monocyclic heptapeptide containing five D-amino acids and two-variable L-amino acids and has a very short life in nature (Pal et al. 2018). It has 'half-life of 90-120 days in natural conditions' (Welker and Steinberg 2000). Ninety-five percent (95%) of Microcystin toxin produced by algal bloom cells are retained within healthy cells (intracellular) during the growth phase of algal blooms. During the senescence and lysis period of the algal bloom, these intracellular microcystins are released into the surrounding (extracellular) environment, resulting in a higher concentration of MCs in the water bodies (Zhang et al. 2009). This toxin can also be released during drinking water treatment process (McLellan and Manderville 2017).

Microcystin was first isolated from *Microcystis aeruginosa*, but it is also produced by other algal bloom species such as '*Anabaena*, *Planktothrix*, *Nostoc*, and *Oscillatoria*, etc.' *M. aeruginosa* is most commonly known to be associated with harmful algal blooms. Drinking water supply, in combination with the global rise in temperature, had emerged as the biggest threat to the population living in developing countries like India.

Recently, India has witnessed several extreme events leading to the capital loss of around \$100 billion and affecting life of approx. 300 million citizens. Moreover, rise in algal bloom events will put additional pressure on governments and public agencies regarding the supply of drinking water (Goyal and Surampalli 2018). Sangolkar et al. 2009, reported the presence of *Microcystis aeruginosa* bloom in Ambazari lake of Nagpur, Maharashtra and studied the toxicity of the Lake water. They isolated different cyanobacteria and algae and checked the level of toxin and different variants of toxin they produce. Wells et al. in 2015, speculated that both the frequency and severity of algal bloom would increase with respect to climate change. The knowledge of factors that mediates algal bloom formation is important for both its prediction and mitigation in the coming future (Wells et al. 2015). Previously, very few studies have been done on the prevalence and consequences of algal blooms in India. In 2010, a Swedish group belonging to the University of Gothenburg proposed that the epidemic of algal blooms also affects the food access and livelihood of poor people in India. According to their study, a significant fraction of the Indian population depends on the sea as the source of

food and income. In aquaculture industry, antibiotics are also used in subtherapeutic concentrations to avoid bacterial/viral contamination, and this leads to the evolution and dissemination of antimicrobial resistance.

Further, in cities, treated sewage is also being released in these water sources, which leads to organic enrichment like nitrogen, phosphorous and enhances the concentration of antibiotic-resistant genes (ARGs), antibiotic-resistant bacteria (ARBs). Organic enrichment increases the episodes of algal blooms. It plays a significant role in the widespread antimicrobial resistance posing a threat to the populations. Few lakes were already studied for the presence of Harmful Algal Blooms and presence of toxins in Nagpur city. Therefore, in the present study, we have explored the taxonomic, catabolic and resistome profile of three freshwater reserves situated around the City, i.e. Ambazari (MP_AQ_350), Gorewada (GW_350) and Gandhisagar (GS_350) using metagenomic approach to validate previous study. These three lakes were chosen because they serve as primary sources of drinking water to the Nagpur citizens and because of their similar landscape. This is the first study aiming at microbiome profiling of algal bloom contaminated lake water from India. Comparative analysis was also done for all three metagenomes concerning cyanobacterial abundance in terms of microbiome associated with and without algal bloom.

Metagenomics is a next-generation tool for microbial ecologists, which provides answers to the key question: who is there? Metagenomics has the ability to unlock the BlackBox of microbial ecosystems. Metagenomics also has an advantage over conventional microbiological/biotechnological approaches that it is fast, rapid and provides a large amount of data, whereas the latter is laborious and time-consuming. We have also analyzed the presence of microcystin toxins in all samples using microcystin gene-specific PCR.

2. Materials and methods

2.1 Sampling, metagenomic DNA extraction and quantification

Algal bloom samples were collected from three different lakes, i.e. Gorewada lake (GW_350), Ambazari Lake (MP_AQ_350), and Gandhinagar Lake (GS_350) situated around Nagpur city, Maharashtra, India (supplementary figure 1). Samples were

collected from each lake in triplicates (500 ml sampling bottles) for three seasons; pre-monsoon, post-monsoon and monsoon. After sampling, the samples were immediately preserved in dry ice and were transferred to the lab. Physiological parameters were checked using automated probes that are in the lab, which measures pH, temperature, turbidity, conductivity. BOD, COD, nitrates, total phosphates, Ammonical nitrogen, total suspended solids, and total dissolved solids were calculated using APHA protocols (table 1; only pre-monsoon data shown). Quantitative analysis of phytoplankton was done using Sedgwick- Rafter cell method (McAlice 1971). Bloom samples were diluted for all seasonal samples and then calculated. In this study, Metagenome analysis was done only for pre-monsoon (summer) samples, as these samples were observed with higher bloom occurrence. Samples were centrifuged at 10000 rpm and palette was used for DNA isolation. Total DNA was extracted using Fast DNA Spin Kit for Soil, MP Biomedicals. The presence of DNA was determined by running agarose gel (1%) electrophoresis at 140 V for 45 min. Purity check (A260/280 ratio) of DNA was done using Denovix Nanodrop (model DS-11). Also, Qubit 2.0 fluorometer was used for accurate measurement of DNA concentration. After confirmation, metagenomic sequencing was done using the Illumina platform after library clustering with paired-end reads (Ye *et al.* 2016).

2.2 Metagenomic analysis using MG-RAST, STAMP, BusyBee and Jvenn

The metagenome of all three lake water samples was uploaded on 'MG-RAST database' and was analyzed using default parameter. It is a publicly available database that allows the mapping of catabolic and taxonomic potential of environmental metagenome. We also analyzed the presence of antibiotic and metal resistance genes present in all three-lake water using MGRAST. The annotation of metagenomic sequences was shown in table 2. Statistical analysis of metagenomes was carried out by Statistical Analysis of Metagenomic Profiles (STAMP) software (Parks *et al.* 2014, Yadav and Kapley 2019). Also, Jvenn tool was used for the representation of common species (up to genus level) present in all the metagenomic samples (Bardou *et al.* 2014). BusyBee is also used, which is a web-based tool available for the metagenomic analysis of NGS data. It can be used for both taxonomic and functional profiling (ARGs annotation) of metagenome. It requires input sequences in the FASTA format. BusyBee allows automatic annotation of assembled contigs into population-level genomic bins and ARGs sequences using bootstrap and unsupervised approach. The output results are represented in the form of 2D scatter plot. Individual dots represent single sequences, and ARGs sequences are represented as black dots (Laczny *et al.* 2017).

Table 1. Water quality parameters of three lakes (Ambazari MP.AQ_350, Gorewada_350, and Gandhi Sagar GS-350)

Parameters	Ambazari MP.AQ_350	Gorewada GW_350	Gandhi Sagar GS_350
Latitude	21°7'42.32'N	21°10'60.40' N	21°14'64.21' N
Longitude	79°2'39.74'E	79°01'60.62' E	79°09'77.04' E
Depth	5m	3m	2.5m
pH	7.03	7.19	7.21
Temperature(°C)	37.2	33.8	36.6
Turbidity (NTU)	7	5	6
Total Suspended Solids (mg/L)	30	36	33
Total dissolves solids (mg/L)	285	292	287
Conductivity (ms/cm)	483	479	491
Nitrate as NO (mg/L)	0.45	0.35	0.39
Total phosphate (mg/L)	0.51	0.39	0.41
Ammonical Nitrogen (mg/L)	ND	ND	1.12
DO (mg/L)	4.9	5.7	5.4
COD (mg/L)	87	68	64
BOD (mg/L)	8.2	7	7.9

2.3 HPLC, PCR based analysis of microcystin toxin and ARGs

DNA isolated from all three sampling locations were subjected to microcystin gene-specific PCR. The primers used for PCR are listed in supplementary table 1 (Li and Zhang 2020). The amplification profile for the reaction was, 94°C for 3 min, 94°C for 45 s for 35 cycles, annealing temperature at 58°C for 1 minute, 72°C for 2 minutes, 72°C for 7 minutes followed by cooling at 4°C. The PCR products were checked by running on 1.5 % agarose gel stained with ethidium bromide. The presence of microcystin (*mcyA*) was also confirmed using HPLC (Perkin Elmer Shelton CT 06484 USA) analysis. Lake water samples were sonicated and filtered using 0.2 micron filter paper. 20 microliters of supernatant were injected into UV detector HPLC using C-18 column with 95:5 acetonitrile and water gradient and flow rate of 1 mL/minute.

3. Results and discussion: Metagenomic analysis of lake water metagenome

3.1 Taxonomic profiling of lake water metagenomes

Prokaryotes are natural engineers of ecosystems. They act as drivers of biogeochemical processes (for example, bioremediation climate change, and public health) undergoing in the environment like freshwater lakes. Biogeochemical processes (anoxic or hypoxic) undergoing in ecosystems are often affected by both local prevailing conditions, as well as global geographic trends. Understanding the correlation between biogeochemical processes response concerning global climate change will help in designing future bioremediation strategies ensuring public health and ecosystem services. Oxygen depletion is driven by eutrophication of aquatic systems affecting microbial diversity and economically important marine organisms (Fire *et al.* 2020; Sun *et al.* 2018). Lake Ambazari and Gorewada water were impacted with algal blooms, whereas in Gandhisagar algal bloom was not visible. Therefore, we have analyzed the metagenomes of all three lakes for understanding the community profile associated with and without algal blooms. It was observed that *Microcystis aeruginosa* was dominant in Ambazari lake species, and therefore it can be considered as Harmful Algal Bloom (HAB) due to microcystin production. Also, it occurs due to single species dominance (Maske *et al.* 2010). In Gorewada and Gandhi

Table 2. Annotation of metagenomic datasets obtained from lake water (Ambazari lake MPAQ_350, Gorewada lake GW_350 and Gandhi Sagar lake GS-350) using MG-RAST

Feature	Ambazari MPAQ_350	Gorewada GW_350	Gandhi Sagar GS_350
Total sequences	28,918,238	31,083,652	27,842,276
Average read length (bases)	150 ± 1	150 ± 1	150 ± 1
Post QC sequence	24,287,279	25,347,346	23,708,733
Average GC content	55 ± 13 %	61 ± 12 %	56 ± 13 %
QC passed reads	53,792	32,480	45,718
Predicted protein feature	20,266,761	23,473,270	21,704,733
Predicted rRNA features	84,325	63,600	89,359
Identified protein features	6,367,317	8,324,368	7,143,484
Identified rRNA features	13,500	14,418	21,260

Sagar lake, mixed species were observed. Potentially resuspended cells (PR cells/ml) approach was used for the quantification of algal bloom (table 3).

Metagenomic profiling indicated positive correlation between water quality parameters and prevalence of algal blooms. In Ambazari lake the levels of phosphates, nitrates, BOD, and COD were higher in concentration, resulting in higher bloom growth (table 1). Also, DO level was found lesser in Ambazari lake. Salinity of lake water under study was <1%, however reports have shown that, for occurrence of algal bloom conditions salinity should be between 23 and 28‰ (Yang 2008). Although, salinity of lake water observed was very low, but it did not have significant impact on the episodes of algal bloom under prevailing conditions (table 1). Algae comes under the category of r-strategist organism as they can grow rapidly and have shorter life span. Temporal variations of three lake water samples were seen with respect to Potentially Resuspended cells (PR cells/ml) and other parameters (tables 1 and 3). The water temperature at the time of bloom (summer) was 32.5°C in Ambazari lake whereas in Gorewada and Gandhisagar lake it was 26°C and 27°C. It has been reported that temperature around 25°C or higher is required for growth of bloom-forming cyanobacteria (Maske *et al.* 2010). Hence, temperature of 32.5°C

Table 3. Temporal variation in Potential Resuspended cells per ml (PR cells ml⁻¹) in water from three lakes (Ambazari MP.AQ_350, Gorewada_350 and Gandhi Sagar GS-350)

Sampling Site	Pre-monsoon (PR cells ml ⁻¹)	Monsoon (PR cells ml ⁻¹)	Post-monsoon (PR cells ml ⁻¹)
Ambazari Lake MP.AQ_350	1.23×10^8	1.39×10^6	1.41×10^7
Gorewada Lake GW_350	1.05×10^4	1.25×10^4	1.85×10^4
Gandhisagar Lake GS_350	1.43×10^3	1.13×10^3	1.93×10^3

allowed proliferation of algal blooms in Ambazari lake than other two lakes under study. With respect to effect of nutrients on understanding r- and k- strategy, algal blooms showed dominancy indicating them as r-strategists. Because of high nutrients in lakes and the presence of single dominating species, lakes showed selection of r-strategists and low diversity. It has been also reported that phytoplankton succession mainly depends upon nutrient availability and fast growth, so occurrence of r-strategists could be predicted, which is characterized by small cell sizes and high maximum growth rates (Wentzky 2020). In present study, temperature and high nutrients were found contributing for the rise of algal blooms (as r-strategist) conditions in Ambazari lake during summer season.

Phylogenetic comparison of lake water metagenome was carried out at multiple taxonomic levels. Metagenomic analysis confirmed that all three lake metagenomes harbour huge prokaryotic diversity. Despite enormous diversity, 990 bacterial species were common in all three metagenomes (figure 1). Similarly, 25 genera were common in Ambazari and Gorewada samples, whereas 43 genera were common in Ambazari and Gandhisagar and 40 genera were common between Gorewada and Gandhisagar samples.

At domain level, bacteria were most abundant in all the metagenomes comprising 97.01%, 98.30%, and 95.90% in Ambazari, Gorewada, and Gandhisagar, respectively (table 4). Among the phylum level, Proteobacteria were most abundant, followed by Cyanobacteria and Bacteroidetes in all the metagenomes. Since cyanobacteria are responsible for algal bloom formation, specifically their abundance were analyzed in all the samples. It was found that cyanobacterial population, especially *Microcystis* abundance, was highest in Ambazari lake (figure 2).

Microcystis sp. is well known for the production of cyanotoxins i.e., microcystin, which is hepatotoxic and neurotoxic in nature (Maske *et al.* 2010). The orders ‘*Oscillatoriales*, *Nostocales*, and *Chroococcales*,’ were most abundant in all the three metagenomes. All three orders consist of those organisms which release microcystins into the environment (Grabowska and Pawlik-Skowrońska 2008). Also, the prevalence of nitrogen-fixing cyanobacteria ‘*Chroococcales*’ in all three samples was observed. Various other bacteria responsible for waterborne diseases such as *Vibrio cholera*, *Pseudomonas*, *Campylobacter jejuni*, *Shigella flexneri* etc were present in all three lake water samples. Metagenomic analysis also indicated the presence of *Trichodesmium* ‘marine nitrogen-fixing cyanobacteria’ in all three lake water samples. They are also responsible for toxin production (Bif *et al.* 2019; Proença *et al.* 2009). Archaeal abundance was highest in Gandhisagar (2.22%) followed by Ambazari (1.43%) and Gorewada (0.84%) samples, respectively. Recent studies have found the occurrence of methanogenic archaea in association with *Microcystis* sp. during an algal bloom. In this association, cyanobacteria (aerobes) provides a protective function to the methanogenic archaea (anaerobes) (Batista *et al.* 2019).

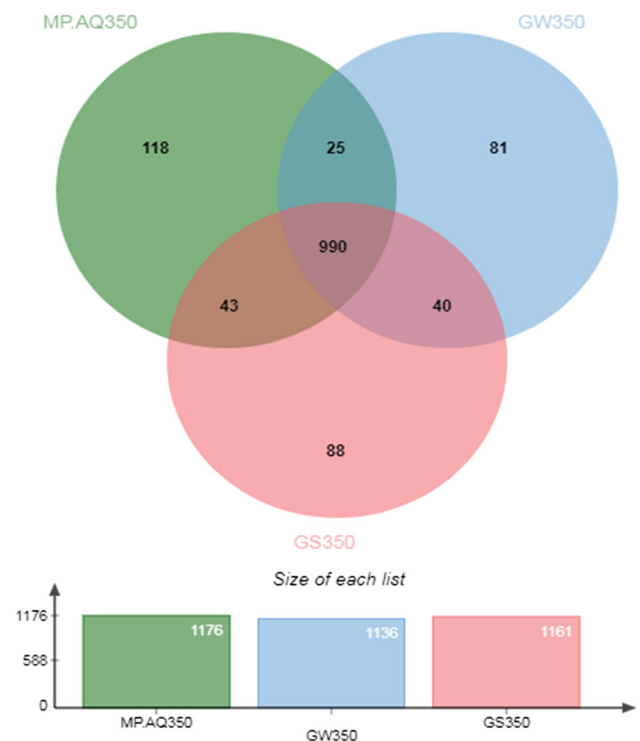
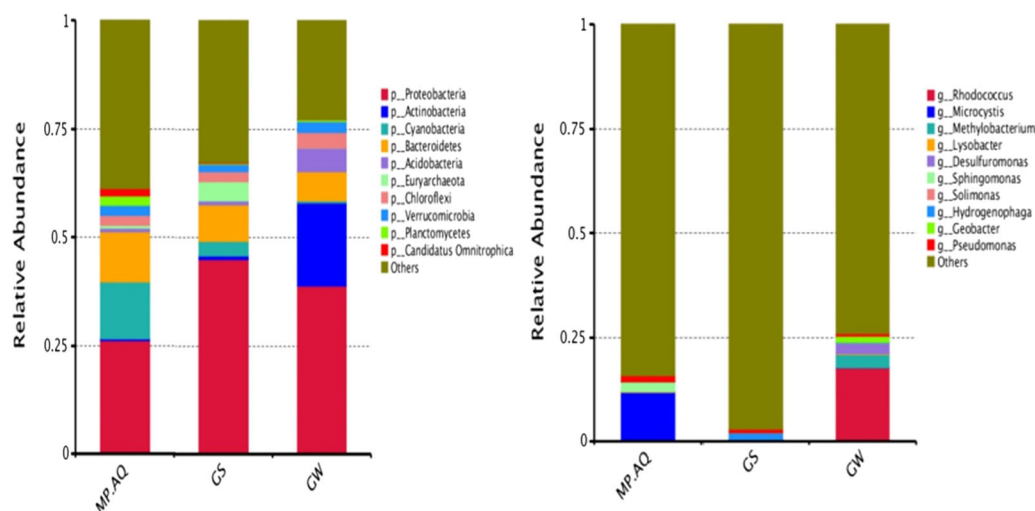
**Figure 1.** Venn diagram of shared taxa (up to genus level) of three metagenomes obtained from the study lakes (Ambazari MP.AQ_350, Gorewada_350, and Gandhi Sagar GS-350).

Table 4. Domain level distribution of microbiome associated with three lakes (Ambazari MP.AQ_350, Gorewada_350 and Gandhi Sagar GS-350)

Domain	Ambazari MP.AQ_350	GorewadaGW_350	Gandhi SagarGS_350
Bacteria	97.01	98.30	95.90
Archaea	1.43	0.84	2.22
Eukaryotes	1.27	0.59	1.51
Viruses	0.12	0.04	0.16
Unclassified sequences	0.17	0.22	0.16
Other	0.0	0.0	0.0

**Figure 2.** Taxonomic composition associated with water from three lakes (Ambazari MP.AQ_350, Gorewada_350, and Gandhi Sagar GS-350) at genus and phylum level.

3.2 Functional profiling of lake water metagenomes

The presence of various pathways (up to level 1) in metagenomes were statistically compared using STAMP software. The Ambazari metagenome was compared with Gorewada and Gandhisagar metagenome. In total, 29 catabolic features were compared. This includes genes involved in DNA and RNA metabolism, respiration, virulence, disease and defence, cofactors, vitamins, prosthetic groups, pigments, membrane transport, etc. 23 functions showed significant association (95% confidence interval) between Ambazari and Gandhisagar metagenomes, whereas 26 functions showed significant association (95% confidence interval) between Ambazari and Gorewada (supplementary figure 2a, b). It may be concluded that despite similar community profiles in all the three metagenomes (990 shared genus), the predominant pathway may vary with the samples. About 99% of microbes elude culturing, and what is their role in ecosystem function is yet to be discovered. This

depicts the huge knowledge gap associated with microbial world. Metagenomic analysis may shed light on microbiome associated with the previously under-explored niches. As evident with the taxonomic profile analysis, 990 genera were common in all the lakewater metagenomes. They encode for the predominant pathway and nutrient recycling in such habitats (Steffen *et al.* 2012). Toxicity of algal bloom is linked with microcystin production because of its hepatotoxic and neurotoxic in nature. *Microcystis aeruginosa* is responsible for microcystin production during algal bloom (Saraf *et al.* 2018). Despite *Microcystis* presence, *mcyA* gene was not detected in water metagenomes, but microcystin dependent protein (*mdpB*) was found. *mdpB* protein has been poorly characterized and its function is not known yet. Other toxins such as Cholera toxin, Diphtheria toxin, Pertussis toxin and *Streptolysin S* Biosynthesis and transport were present in all three lake water metagenomes. Presence of these toxins in drinking water may pose a threat to public health (Schmitt *et al.* 1999). Climate condition also promotes *Microcystis* growth during algal bloom. It has

been found that average optimal temperature for *Microcystis* growth is 20–30°C, and the average temperature of Nagpur city is 26.5°C, which is suitable for the algal bloom to grow (Yang *et al.* 2018). Phylogenetic composition of all three-lake water samples were also compared (supplementary figure 3a and b).

3.3 Prevalence of microcystin toxin in lake water

For confirmation of microcystin production *mcyA* gene was used as a biomarker. Gene-specific PCR was performed to confirm the presence of *mcyA* gene. Axenic *Microcystis* culture (250 ml flask) was used as a positive control. The *mcyA* gene was detected only in positive control and the Ambazari lake water sample (supplementary figure 4). Gorewada and Gandhisagar lake water showed negative amplification. *mcyA* gene was not detected even after metagenomic analysis. Previously, Beversdorf *et al.* (2015) reported that *mcyA* gene cannot be used as biomarker of microcystin concentrations in lakes. Previously, it has been reported that the concentration of microcystin may vary from environmental factors and time, so measuring the gene copy number only cannot explain the fixed sampling point variation (Beversdorf *et al.* 2015). Earlier we have isolated the *Microcystis* culture from Ambazari lake and found the presence of microcystin toxin using HPLC and GC-MS (Pal *et al.* 2018). In the present study, we also analyzed the presence of microcystin using HPLC. The concentration of microcystin toxin in Ambazari lake water sample was 18.75 µg/l. In Gorewada and Gandhisagar lake water sample concentrations of microcystin 1.72 µg/l and 5.05 µg/l respectively. It was found that the microcystin toxin was present in all three lake water samples and its concentration was higher than the permissible limit recommended by WHO (Sangolkar *et al.* 2009).

3.4 Resistome profiling of lake water samples

Antibiotic resistance had emerged as a global threat to the global population. As these freshwater bodies are being used for the supply of drinking water, we explored the types of genes conferring antibiotic and metal resistance. In total 32 types of genes encoding antibiotic and metal resistance (MRGs) were found (supplementary figure 5). Antibiotic resistance genes encoding for vancomycin resistance ‘last resort of antibiotics,’ polymyxin, fluoroquinolones, erythromycin resistance, beta-lactamase, etc. were found in all three lakes water

metagenomes. The presence of beta-lactamase indicates that there is resistance over lactam family antibiotics like cephalosporin, penicillin, carbapenems, and cephamycin (Wang *et al.* 2018). Resistance genes conferring resistance to metals like copper, cobalt, cadmium, zinc, arsenic resistance, etc. were also abundant in all the metagenomes. These metals play an important role in the co-selection of ARGs in the environments (Swift *et al.* 2019; Waseem *et al.* 2017). Efflux pumps like MexA-MexB-OprM_Multidrug_Efflux_System, MexC-MexD – OprJ_Multidrug_Efflux_System, MexE-MexF-OprN_Multidrug_Efflux_System encoding for multiple drug resistance were also found in all the metagenomes (figure 3a, b and c). The presence of multidrug-resistant efflux system indicates the presence of multidrug resistance bacteria, which has primary public concern nowadays; there are very few effective antimicrobial agents present which can treat infections caused by gram-negative bacteria (Guo *et al.* 2017). The finding suggests that lake water harbours complex resistome profiles. Toxins and superantigens profiling were also seen in all three lake water samples (supplementary table 2). These habitats may serve as a hotspot exchange of resistant genes (Yadav and Kapley 2019). Bacteria often acquire resistance by horizontal gene transfer (HGT). The mechanism of acquiring antimicrobial resistance is well known, but how climate change drives the dissemination of antibiotic resistance is yet to be explored (Bengtsson-Palme and Larsson 2015). Recently, Macfadden and coworkers have found that rising temperatures may alter and accelerate the antibiotic resistance phenomenon (MacFadden *et al.* 2018). With the current trends of the rise in antibiotic resistance, the ‘post-antibiotic era’ will soon become a reality of the 21st century (Reardon 2014). A multifaceted global approach is required to combat antibiotic resistance. World Health Organization (WHO) had expressed its concern over AMR stating ‘No action today means no cure tomorrow’. Therefore, the WHO had proposed a ‘global action plan (GAP)’ to combat antimicrobial resistance. (WHO 2014, 2017). BRIC countries are being considered as an epicentre for antimicrobial resistance. India has already adopted ‘GAP’ to fight the epidemic of AMR (McArthur and Tsang 2017).

The first prevalence of ARGs encoding for tetracycline (tet), chloroamphenicol (CMP) resistance, methicillin resistance, aminoglycoside (aph), sulphonamide (sul1), quinoline group (qnrS), beta-lactamase (bla), microcystin biosynthesis genes (*mcyA* and *E*), and multiple efflux pump (*mexA*) were checked using PCR. The obtained result is in congruence with the recently published article where authors have evaluated



Figure 3. Resistome profiling of three lakes: (a) Ambazari lake, (b) Gorewada lake, (c) Gandhisagar lake.

the abundance of ARGs in water samples with bloom and without bloom (Guo *et al.* 2018). The authors have found that the ARGs richness was lower during bloom conditions. However, it was found that the presence of tetracycline may inhibit cell growth and protein synthesis and induce oxidative stress in *M. aeruginosa*. It was proposed that the optimum concentration of tetracycline can be used for controlling algal bloom episodes in an aquatic environment (Kovalakova *et al.* 2020; Shang *et al.* 2015). MGRast based analysis cannot be used for species-level prediction because of short read lengths. Therefore, BusyBee tool was used for annotation of assembled contigs up to species level and with antibiotic resistance genes. Busybee uses Kraken and Prokka for taxonomic and resistome profiling. Busybee based analysis indicated presence of 744 species, 404 species, and 399 species in Ambazari, Gandhisagar, and Gorewada lakewater samples, respectively. Busybee classified Ambazari contigs into 49 bins indicating taxonomic abundance, which was highest followed by Gandhisagar (15 Bins) and Gorewada (6 Bins) samples, respectively. ‘ESKAPE Pathogens’ like *Acinetobacter baumannii*, *Enterobacter*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and *Vibrio cholera*, etc. were also detected in the lakewater metagenome using BusyBee. *Microcystis aeruginosa* responsible for microcystin production was also found after analysis with BusyBee. Busybee annotated 122

resistance genes in Ambazari, 116 in Gandhisagar, and 122 in Gorewada lake water samples (supplementary figure 6a, b, c). Few additional antibiotic resistance genes, such as tetracycline, were found in the metagenomes. Plasmid, such as Plasmid pSB102, Plasmid F, Plasmid pGNB2, may play a role in HGT transfer of antimicrobial resistance genes (Founou *et al.* 2017; Tommasi *et al.* 2015).

4. Conclusions

Events of algal bloom are rising with climate change. In the present study, genomic tools such as PCR and metagenomics along with HPLC were used for the exploration of microbiome and resistome profile associated with cyanobacterial bloom. Obtained results indicated that all three-lake water encompasses complex taxonomic and catabolic (resistome profile). Various, waterborne pathogens like *Vibrio cholera*, *Pseudomonas*, *Campylobacter jejuni*, *Shigella flexneri* etc. were detected. PCR results confirmed the presence of microcystin (neurotoxic) toxin. The presence of antibiotic resistance genes (ARGs) with microcystin toxin could pose threat to the receiving human population of any city. The present study envisaged the impact of algal blooms on composition and functional microbiomes of lake water and in future will emphasize

towards the impact of antibiotics on algal bloom episodes and vice versa.

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References

- Bardou P, Mariette J, Escudié F, Djemiel C and Klopp C 2014 Jvenn: an interactive Venn diagram viewer. *BMC Bioinform.* **15** 293
- Batista A, Woodhouse J, Grossart H-P and Giani A 2019 Methanogenic archaea associated to *Microcystis* sp. in field samples and in culture. *Hydrobiologia* **831** 163–172
- Bengtsson-Palme J and Larsson DG 2015 Antibiotic resistance genes in the environment: prioritizing risks. *Nat. Rev. Microbiol.* **13** 396
- Beversdorf LJ, Chaston SD, Miller TR and McMahon KD 2015 Microcystin *mcyA* and *mcyE* gene abundances are not appropriate indicators of microcystin concentrations in lakes. *PLoS One*
- Bif MB, de Souza MS, Costa LDF and Yunes JS 2019 Microplankton community composition associated with aggregations of toxic *Trichodesmium* in the Southwest Atlantic Ocean. *Front. Marine Sci.* **6** 23
- Deng Y, Wu M, Zhang H, Zheng L, Acosta Y and Hsu T-TD 2017 Addressing harmful algal blooms (HABs) impacts with ferrate(VI): Simultaneous removal of algal cells and toxins for drinking water treatment. *Chemosphere* **186** 757–761
- Fire SE, Browning JA, Durden WN and Stolen MK 2020 Comparison of during-bloom and inter-bloom brevetoxin and saxitoxin concentrations in Indian River Lagoon bottlenose dolphins, 2002–2011. *Aquat. Toxicol.* **218** 105371
- Founou RC, Founou LL and Essack SY 2017 Clinical and economic impact of antibiotic resistance in developing countries: a systematic review and meta-analysis. *PLoS One* **12** e0189621
- Goyal MK and Surampalli RY 2018 Impact of climate change on water resources in India. *J. Environ. Engin.* **144** 04018054
- Grabowska M and Pawlik-Skowrońska B 2008 Replacement of chroococcales and nostocales by oscillatoriales caused a significant increase in microcystin concentrations in a dam reservoir. *Oceanol. Hydrobiol. Stud.* **37** 23–33
- Guo J, Li J, Chen H, Bond PL and Yuan Z 2017 Metagenomic analysis reveals wastewater treatment plants as hotspots of antibiotic resistance genes and mobile genetic elements. *Water Res.* **123** 468–478
- Guo Y, Liu M, Liu L, Liu X, Chen H and Yang J 2018 The antibiotic resistome of free-living and particle-attached bacteria under a reservoir cyanobacterial bloom. *Environ. Int.* **117** 107–115
- Hofer U 2013 Getting to the bottom of cyanobacteria. *Nat. Rev. Microbiol.* **11** 818–819
- Huisman J, Codd GA, Paerl HW, Ibelings BW, Verspagen JMH and Visser PM 2018 Cyanobacterial blooms. *Nat. Rev. Microbiol.* **16** 471–483
- Kovalakova P, Cizmas L, McDonald TJ, Marsalek B, Feng M and Sharma VK 2020 Occurrence and toxicity of antibiotics in the aquatic environment: a review. *Chemosphere* **251** 126351
- Lacny CC, Kiefer C, Galata V, Fehlmann T, Backes C and Keller A 2017 BusyBee Web: metagenomic data analysis by bootstrapped supervised binning and annotation. *Nucleic Acids Res.* **45** W171–W179
- Li Q and Zhang Q 2020 Prevalence and pollution characteristics of antibiotic resistant genes in one high anthropogenically-impacted river. *PLoS One* **15** e0231128
- MacFadden DR, McGough SF, Fisman D, Santillana M and Brownstein JS 2018 Antibiotic resistance increases with local temperature. *Nat. Clim. Chang.* **8** 510–514
- Maske SS, Sangolkar LN and Chakrabarti T 2010 Temporal variation in density and diversity of cyanobacteria and cyanotoxins in lakes at Nagpur (Maharashtra State) India. *Environ. Monitor. Assessment* **169** 299–308
- McAlice BJ 1971 Phytoplankton sampling with the Sedgwick-Rafter Cell 1. *Limnol. Oceanogr.* **16** 19–28
- McArthur AG and Tsang KK 2017 Antimicrobial resistance surveillance in the genomic age. *Ann. NY Acad. Sci.* **1388** 78–91
- McLellan NL and Manderville RA 2017 Toxic mechanisms of microcystins in mammals. *Toxicol. Res.* **6** 391–405
- Pal M, Pal S, Qureshi A and Sangolkar L 2018 Perspective of cyanobacterial harmful algal bloom (HAB) mitigation: Microcystis toxin degradation by bacterial consortia. *Indian J. Exp. Biol.* **56** 511–518
- Pal M, Yesankar PJ, Dwivedi A and Qureshi A 2020 Biotic control of harmful algal blooms (HABs): A brief review. *J. Environ. Manag.* **268** 110687
- Pal M, Purohit H and Qureshi A 2021 Genomic insight for Algicidal activity in *Rhizobium* sp. (AQ_MP). *Arch. Microbiol.* **203** 5193–5203
- Parks DH, Tyson GW, Hugenholtz P and Beiko RG 2014 STAMP: statistical analysis of taxonomic and functional profiles. *Bioinformatics* **30** 3123–3124
- Proença L, Tamanaha M and Fonseca R 2009 Screening the toxicity and toxin content of blooms of the cyanobacterium *Trichodesmium erythraeum* (Ehrenberg) in

- northeast Brasil. *J. Venom. Animals Toxins Trop. Dis.* **15** 204–215
- Reardon S 2014 Antibiotic resistance sweeping developing world: bacteria are increasingly dodging extermination as drug availability outpaces regulation. *Nature* **509** 141–143
- Riebesell U, Aberle-Malzahn N, Achterberg EP, et al. 2018 Toxic algal bloom induced by ocean acidification disrupts the pelagic food web. *Nat. Clim. Chang.* **8** 1082–1086
- Sangolkar LN, Maske SS, Muthal PL, Kashyap SM and Chakrabarti T 2009 Isolation and characterization of microcystin producing *Microcystis* from a Central Indian water bloom. *Harmful Algae* **8** 674–684
- Saraf SR, Frenkel A, Harke MJ, Jankowiak JG, Gobler CJ and McElroy AE 2018 Effects of *Microcystis* on development of early life stage Japanese medaka (*Oryzias latipes*): Comparative toxicity of natural blooms, cultured *Microcystis* and microcystin-LR. *Aquat. Toxicol.* **194** 18–26
- Schmitt CK, Meysick KC and O'Brien AD 1999 Bacterial toxins: friends or foes? *Emerg. Infect. Dis.* **5** 224
- Shang AH, Ye J, Chen DH, Lu XX, Lu HD, Liu CN and Wang LM 2015 Physiological effects of tetracycline antibiotic pollutants on non-target aquatic *Microcystis aeruginosa*. *J. Environ. Sci. Health B* **50** 809–818
- Steffen MM, Li Z, Effler TC, Hauser LJ, Boyer GL and Wilhelm SW 2012 Comparative metagenomics of toxic freshwater cyanobacteria bloom communities on two continents. *PLoS ONE*
- Sun R, Sun P, Zhang J, Esquivel-Elizondo S and Wu Y 2018 Microorganisms-based methods for harmful algal blooms control: a review. *Bioresource Technol.* **248** 12–20
- Swift BMC, Bennett M, Waller K, et al. 2019 Anthropogenic environmental drivers of antimicrobial resistance in wildlife. *Sci. Total Environ.* **649** 12–20
- Tommasi R, Brown DG, Walkup GK, Manchester JI and Miller AA 2015 ESKAPEing the labyrinth of antibacterial discovery. *Nat. Rev. Drug Discov.* **14** 529–542
- Wang J-H, Lu J, Zhang Y-X, Wu J, Luo Y and Liu H 2018 Metagenomic analysis of antibiotic resistance genes in coastal industrial mariculture systems. *Bioresource Technol.* **253** 235–243
- Waseem H, Williams MR, Stedtfeld RD and Hashsham SA 2017 Antimicrobial resistance in the environment. *Water Environ. Res.* **89** 921–941
- Welker M and Steinberg C 2000 Rates of humic substance photosensitized degradation of microcystin-LR in natural waters. *Environ. Sci. Technol.* **34** 3415–3419
- Wells ML, Trainer VL, Smayda TJ, et al. 2015 Harmful algal blooms and climate change: learning from the past and present to forecast the future. *Harmful Algae* **49** 68–93
- Wentzky VC, Tittel J, Jäger CG, Bruggeman J and Rinke K 2020 Seasonal succession of functional traits in phytoplankton communities and their interaction with trophic state. *J. Ecol.* **108** 1649–1663
- WHO 2014 *Antimicrobial resistance: 2014 global report on surveillance* (World Health Organization)
- WHO 2017 *Global action plan on antimicrobial resistance 2015* (Geneva: World Health Organization)
- Yadav S and Kapley A 2019 Exploration of activated sludge resistome using metagenomics. *Sci. Total Environ.* **692** 1155–1164
- Yang J, Tang H, Zhang X, Zhu X, Huang Y and Yang Z 2018 High temperature and pH favor *Microcystis aeruginosa* to outcompete *Scenedesmus obliquus*. *Environ. Sci. Pollut. Res.* **25** 4794–4802
- Yang XE, Wu X, Hao HL and He ZL 2008 Mechanisms and assessment of water eutrophication. *J. Zhejiang Univ. Sci. B* **9** 197–209
- Ye J, Song Z, Wang L and Zhu J 2016 Metagenomic analysis of microbiota structure evolution in phytoremediation of a swine lagoon wastewater. *Biores. Technol.* **219** 439–444
- Zhang D, Xie P, Liu Y, Chen J and Wen Z 2009 Spatial and temporal variations of microcystins in hepatopancreas of a freshwater snail from Lake Taihu. *Ecotoxicol. Environ. Safe.* **72** 466–472

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